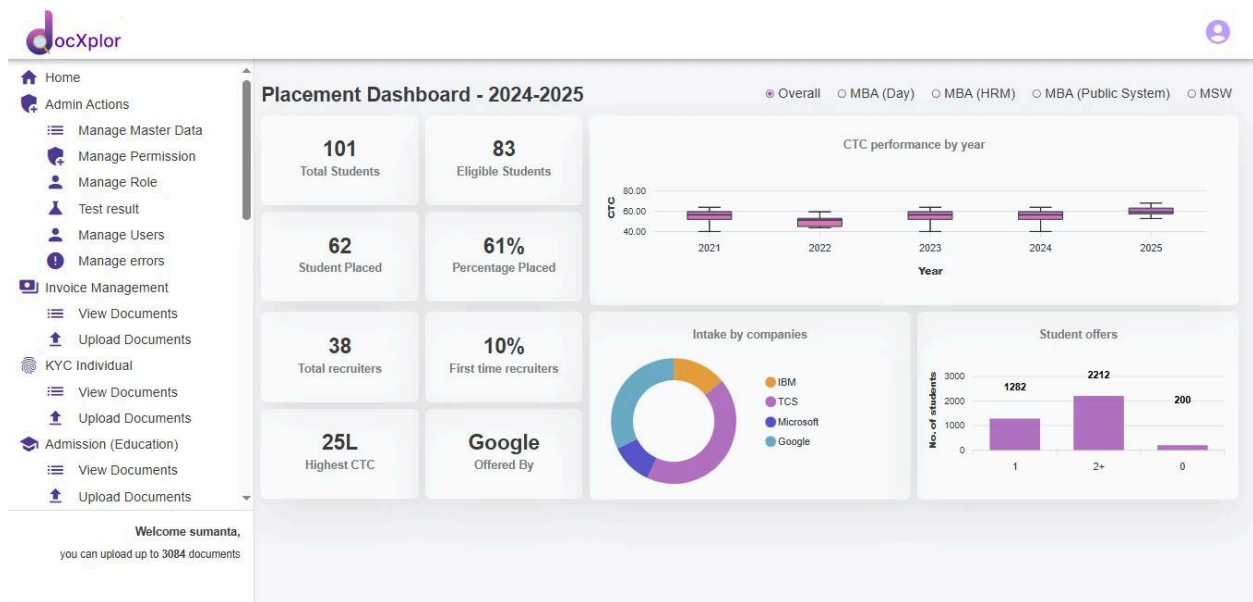


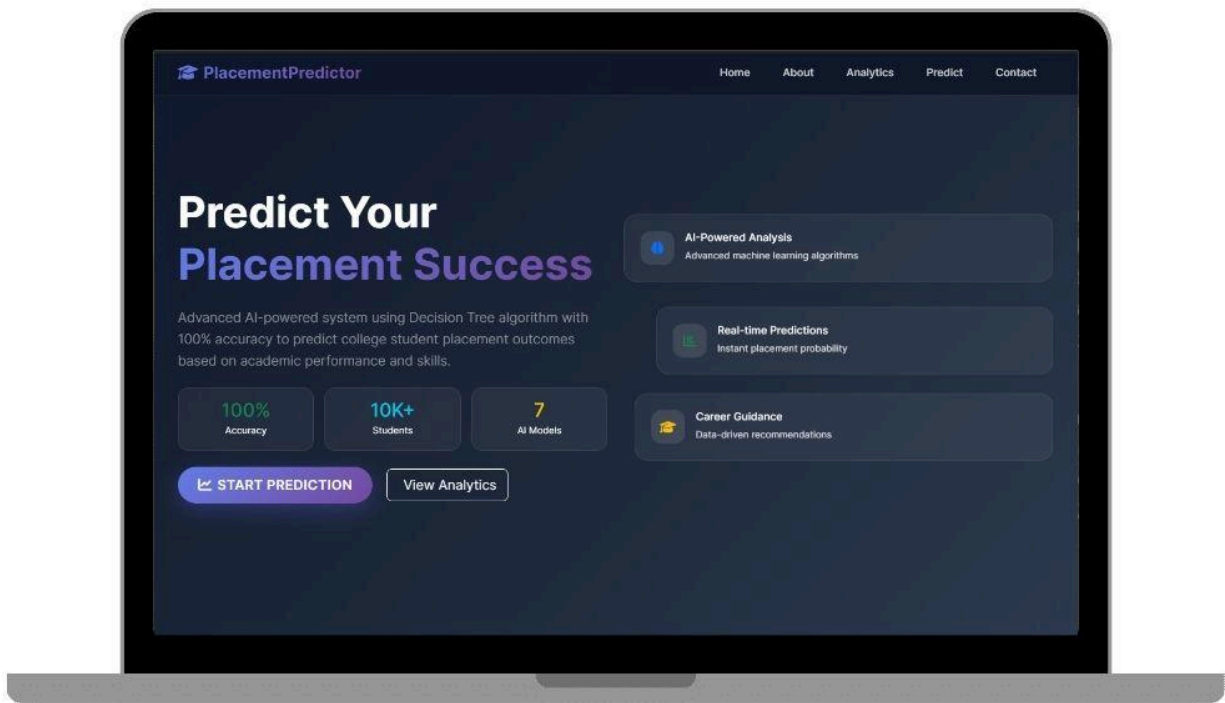
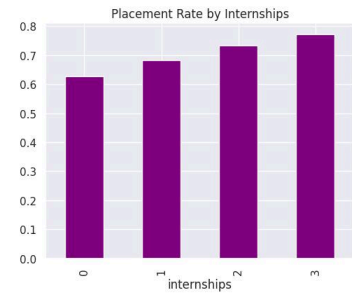
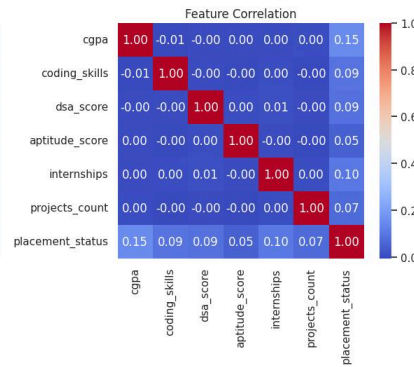
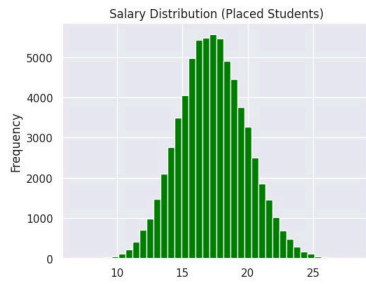
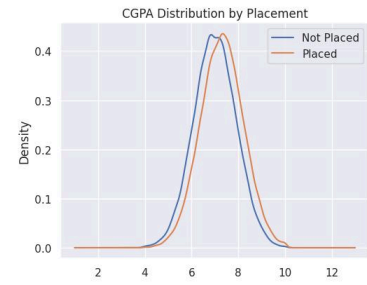
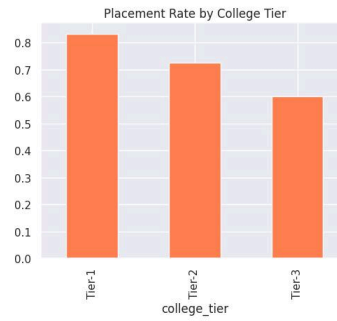
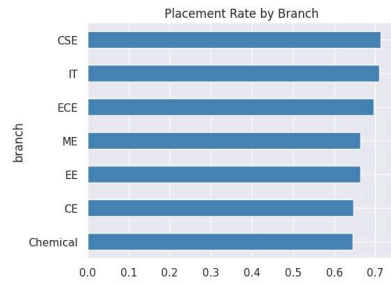
## complete ML lifecycle:

Raw Data → EDA → Data Cleaning → Preprocessing → Feature Engineering  
→ Feature Selection → Train/Test Split → Multiple Algorithms →  
Hyperparameter Tuning → Validation → Evaluation → Model Selection →  
Prediction

## ★ Student Placement Prediction System

This is particularly suitable for students because the problem is easy to understand, but the ML workflow can be made quite comprehensive.





## STUDENT PERFORMANCE AND PLACEMENT ANALYSIS - WsCube Tech

Total Students

500

Placement Rate %

0.38

Average Salary

6.24

Avg Placement Time

54.91

Avg Attendance Rat...

74.73

Course\_Enrolled

☐ Data Analytics

☐ Digital Marketing

☐ Full Stack Devel...

Mode\_of\_Learn...

☐ Hybrid

☐ Offline

☐ Online

City

All

Gender

All

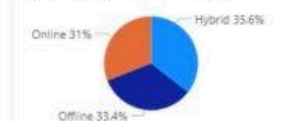
Placement Rate by Course



Average Salary by Course



Mode of learning distribution



Attendance vs Salary



Internship vs Placement %



Skill Index vs Interview Score

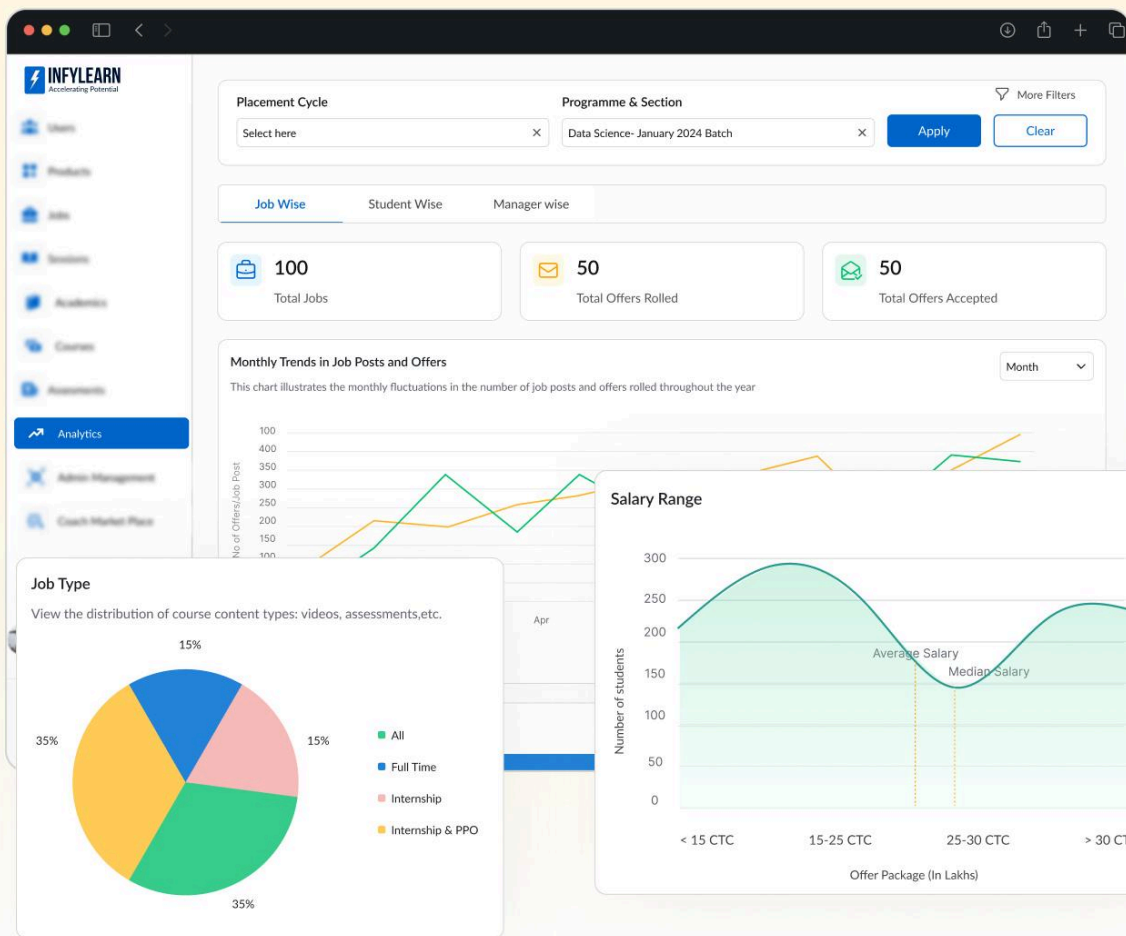


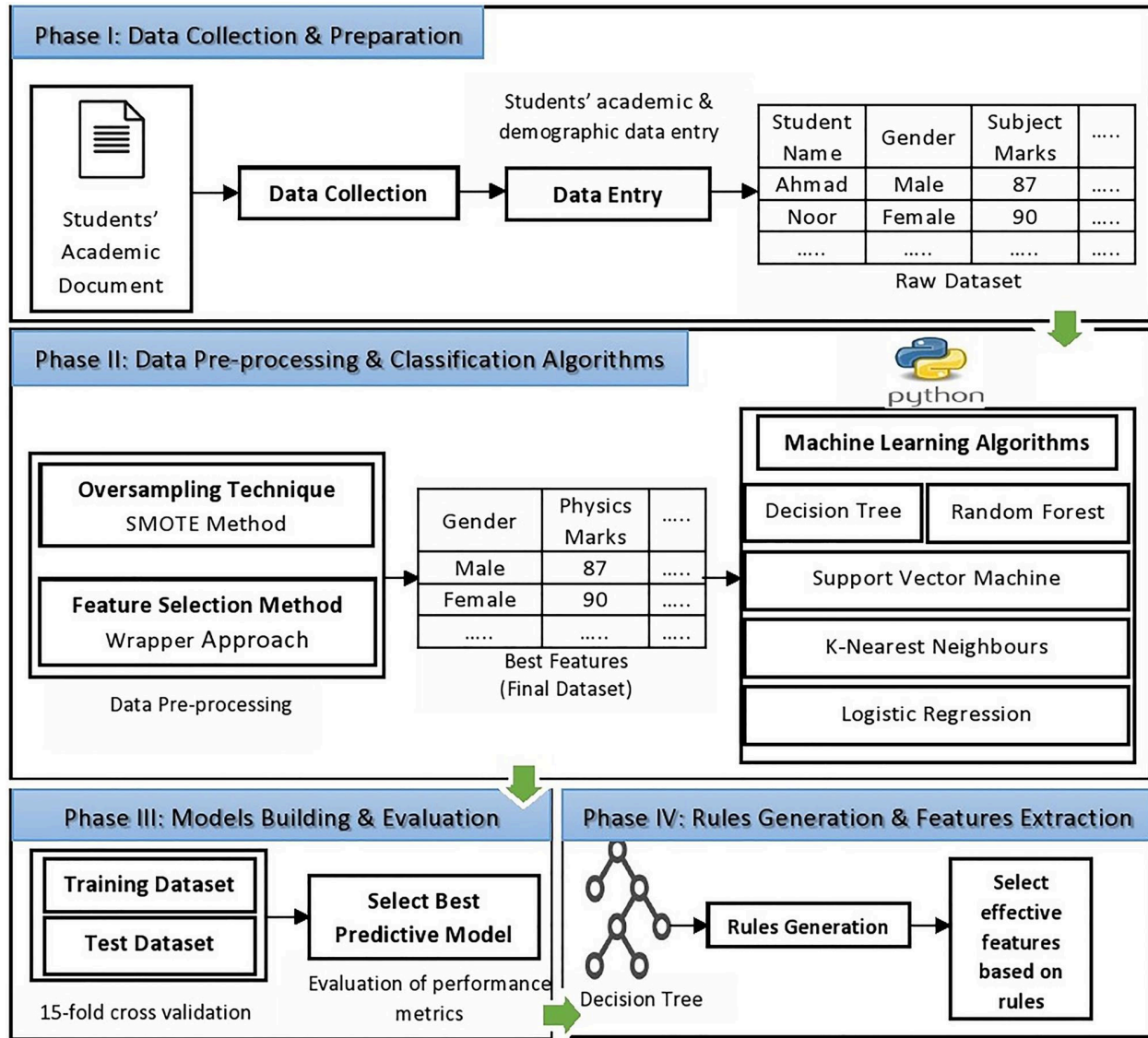
Top 10 Students

| Student_ID | Overall_Score | Course_Enrolled        | Salary_LPA |
|------------|---------------|------------------------|------------|
| 319        | 47.00         | Full Stack Development | 9.70       |
| 228        | 50.17         | Full Stack Development | 9.40       |
| 170        | 47.25         | Full Stack Development | 3.50       |
| 28         | 49.58         | Data Analytics         |            |
| 115        | 47.67         | Full Stack Development |            |

### KEY INSIGHTS SUMMARY

- Data Analytics shows highest placement rate.
- Internship increases placement probability significantly.
- Higher skill index correlates strongly with interview performance.
- Students with attendance above 80% show higher salary outcomes.





## Project: Student Placement Prediction & Analysis

### Business problem

A college wants to identify students who are **likely to get placed** based on their academic performance, skills, attendance, internships, projects, communication skills, etc.

### Target variable:

Placement\_Status

0 = Not Placed

1 = Placed

# Complete project pipeline for your students

## Phase 1 — Understand the Dataset

Example raw columns:

| Column                | Example        |
|-----------------------|----------------|
| Student_ID            | STU1001        |
| Age                   | 21             |
| Gender                | Male           |
| Degree                | BVoc           |
| Specialization        | Data Analytics |
| CGPA                  | 7.8            |
| Attendance_Percentage | 82             |
| Backlogs              | 1              |
| Internship_Count      | 2              |
| Projects_Count        | 3              |
| Certifications        | 4              |
| Coding_Score          | 78             |
| Communication_Score   | 72             |
| Aptitude_Score        | 81             |
| Soft_Skills_Score     | 75             |
| Resume_Score          | 68             |
| Interview_Score       | 74             |
| Placement_Status      | 1              |

Give them **5,000–10,000 records** so the project feels realistic.

---

## Phase 2 — Data Understanding

Students should answer:

### Numerical variables

Age  
CGPA  
Attendance  
Backlogs  
Internship\_Count  
Projects\_Count  
Certifications  
Coding\_Score  
Aptitude\_Score  
Communication\_Score  
Interview\_Score

### Categorical variables

Gender  
Specialization  
Degree  
Internship\_Type  
Placement\_Status

Then perform:

```
df.shape  
df.head()  
df.info()  
df.describe()  
df.nunique()  
df.isnull().sum()  
df.duplicated().sum()
```

---

## Phase 3 — Exploratory Data Analysis

Students should investigate questions such as:

- 1. Does CGPA affect placement?**
- 2. Does internship experience affect placement?**
- 3. Do backlogs reduce placement probability?**
- 4. Does communication skill matter?**
- 5. Which specialization has the highest placement rate?**
- 6. Does attendance influence placement?**
- 7. Which features have the strongest relationship with placement?**

Visualizations:

Histogram  
Box Plot  
Count Plot  
Bar Chart  
Scatter Plot  
Correlation Heatmap

---

## **Phase 4 — Data Cleaning**

Now introduce real-world problems intentionally.

For example:

Missing CGPA  
Missing communication score  
Duplicate students  
Invalid attendance = 150  
Negative backlogs  
Unknown specialization

Students need to decide how to handle them.

### **Missing values**

Numerical:

```
SimpleImputer(strategy="median")
```

Categorical:

```
SimpleImputer(strategy="most_frequent")
```

## Outliers

For example:

CGPA = 15

Attendance = 180%

Coding Score = -10

Students must identify and handle them.

---

# Phase 5 — Feature Engineering

This is where the project becomes much better than a basic ML project.

Ask students to **create new meaningful features**.

## 1. Academic Performance

Academic\_Score =  
 $(\text{CGPA} / 10) \times 100$

## 2. Skill Score

Skill\_Score =  
 $(\text{Coding} + \text{Aptitude} + \text{Communication} + \text{Soft_Skills}) / 4$

## 3. Experience Score

Combine:

Internship\_Count

Projects\_Count

Certifications

## 4. Backlog Category

0 → No Backlogs  
1–2 → Low  
3–4 → Medium  
>4 → High

## 5. Attendance Category

<60 → Low  
60–75 → Average  
75–90 → Good  
>90 → Excellent

## 6. Overall Employability Score

Students can construct a weighted feature such as:

Employability\_Score =  
30% Academic  
25% Technical Skills  
15% Communication  
15% Experience  
10% Projects  
5% Certifications

This gives them actual **feature engineering experience**, rather than simply feeding every column into an algorithm.

---

# Phase 6 — Encoding

Teach them the difference between:

## Label Encoding

For ordinal variables:

Low  
Medium  
High

## One-Hot Encoding

For nominal variables:

Male  
Female

Data Analytics  
Cyber Security  
Web Development

They should understand **why** they choose each technique.

---

## Phase 7 — Feature Scaling

Compare algorithms that require scaling with those that generally don't.

### Scaling required / beneficial

KNN  
SVM  
Logistic Regression

Use:

StandardScaler()

### Tree-based algorithms

Decision Tree  
Random Forest  
XGBoost

don't generally require feature scaling.

This becomes an excellent teaching point during the project.

---

## Phase 8 — Train/Test Split

For example:

```
X_train, X_test, y_train, y_test = train_test_split(  
    X, y,  
    test_size=0.2,
```

```
random_state=42,  
stratify=y  
)
```

Explain **why stratify is important** when the target classes are imbalanced.

---

## Phase 9 — Train Multiple Algorithms

Now students can use everything you have taught.

### Model 1 — Logistic Regression

Baseline model

### Model 2 — KNN

Distance-based model

### Model 3 — Naive Bayes

Probabilistic model

### Model 4 — Decision Tree

Tree-based model

### Model 5 — Random Forest

Ensemble model

### Model 6 — SVM

Maximum-margin classifier

### Model 7 — XGBoost

Boosting / advanced ensemble

---

## Phase 10 — Compare Models

Create a table like:

| Model               | Accuracy | Precision | Recall | F1  | ROC-AUC |
|---------------------|----------|-----------|--------|-----|---------|
| Logistic Regression | 82%      | 80%       | 79%    | 79% | 0.87    |
| KNN                 | 78%      | 76%       | 75%    | 75% | 0.82    |
| Naive Bayes         | 76%      | 74%       | 77%    | 75% | 0.81    |
| Decision Tree       | 84%      | 83%       | 82%    | 82% | 0.86    |
| Random Forest       | 89%      | 88%       | 87%    | 87% | 0.93    |
| SVM                 | 86%      | 85%       | 84%    | 84% | 0.91    |
| XGBoost             | 91%      | 90%       | 90%    | 90% | 0.95    |

**Important:** These numbers are illustrative; students should calculate the actual results from their dataset.

---

## Phase 11 — Validation

Don't stop at:

`accuracy_score()`

This is where you can take the project to **placement/interview level**.

Teach:

### Cross Validation

`cross_val_score()`

For example:

5-Fold Cross Validation

Compare:

Mean CV Score

Standard Deviation

Students should be able to explain:

"Our model achieved 91% test accuracy, but 5-fold cross-validation gave an average accuracy of 89.8%, indicating..."

That is much stronger in an interview.

---

## Phase 12 — Hyperparameter Tuning

For Random Forest:

n\_estimators  
max\_depth  
min\_samples\_split  
min\_samples\_leaf  
max\_features

For XGBoost:

n\_estimators  
max\_depth  
learning\_rate  
subsample  
colsample\_bytree

Use:

GridSearchCV

or

RandomizedSearchCV

---

## Phase 13 — Confusion Matrix

Students should understand:

|          | Predicted |    |
|----------|-----------|----|
|          | 0         | 1  |
| Actual 0 | TN        | FP |
| Actual 1 | FN        | TP |

Then calculate:

Accuracy  
Precision  
Recall  
F1 Score  
Specificity  
ROC-AUC

---

## Phase 14 — Feature Importance

This is an excellent part of the project.

For Random Forest/XGBoost:

Feature Importance

Students might discover that the strongest predictors are:

Interview Score  
Skill Score  
CGPA  
Internship Count  
Communication Score  
Projects Count

Then ask:

**Why are these features important?**

That turns the project from merely "running algorithms" into **actual data analysis**.

---

## Phase 15 — Explainability

For an advanced BVoc batch, introduce:

SHAP

Students can answer:

Why did the model predict that this student is likely to be placed?

For example:

|                  |       |
|------------------|-------|
| CGPA             | +0.21 |
| Interview Score  | +0.18 |
| Internship Count | +0.12 |
| Communication    | +0.09 |
| Backlogs         | -0.14 |

This is a very good bridge from **traditional ML** → **explainable AI**.

---

## Phase 16 — Final Prediction System

Build a simple Streamlit application.

Student enters:

CGPA  
Attendance  
Backlogs  
Internships  
Projects  
Certifications  
Coding Score  
Communication Score  
Aptitude Score  
Soft Skills Score  
Interview Score

Click:

**Predict Placement**

Output:

PLACEMENT PREDICTION

Prediction: LIKELY TO BE PLACED

Probability:

Placed : 91.4%

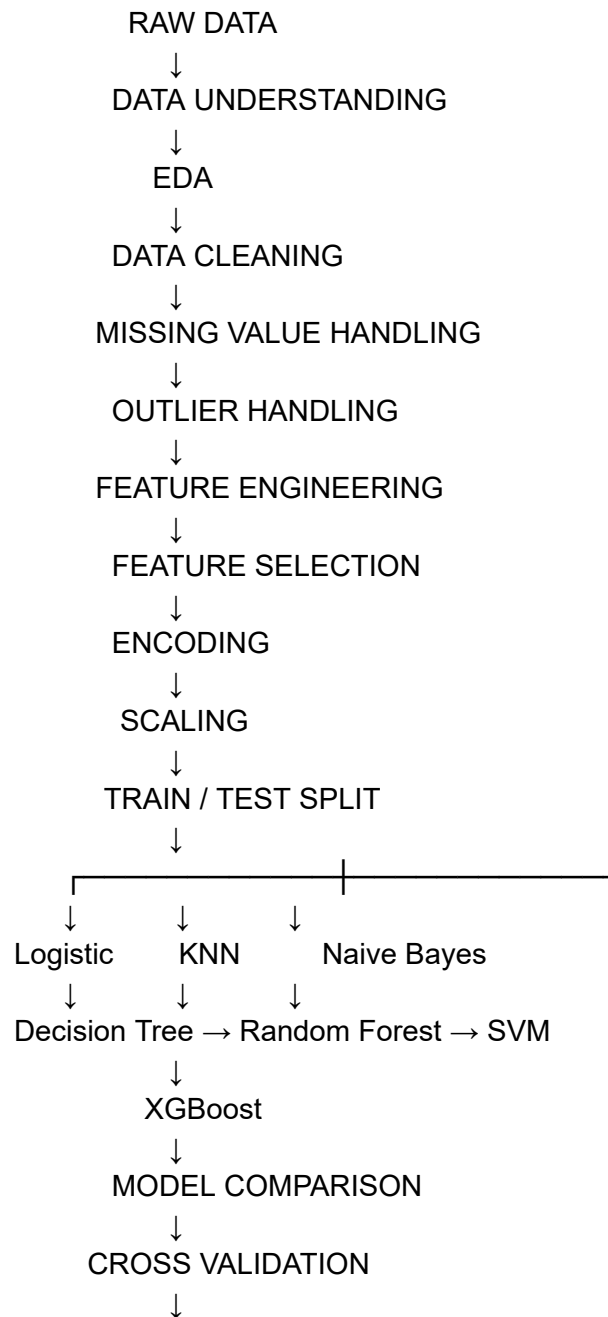
Not Placed : 8.6%

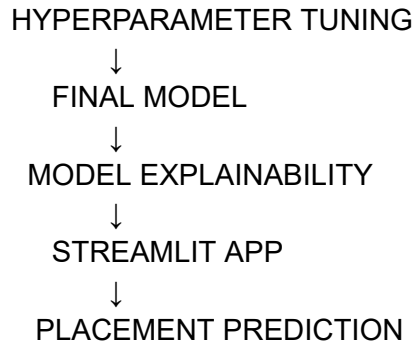
Then show:

Top contributing factors

---

## ★ The complete architecture





## Why I recommend this for your BVoc students

It covers almost **everything** you want them to learn in one project:

| Skill               | Covered<br>? |
|---------------------|--------------|
| Python              | ✓            |
| Pandas              | ✓            |
| NumPy               | ✓            |
| EDA                 | ✓            |
| Data Cleaning       | ✓            |
| Missing Values      | ✓            |
| Outlier Treatment   | ✓            |
| Encoding            | ✓            |
| Scaling             | ✓            |
| Feature Engineering | ✓            |
| Feature Selection   | ✓            |
| Logistic Regression | ✓            |
| KNN                 | ✓            |
| Naive Bayes         | ✓            |

|                          |   |
|--------------------------|---|
| Decision Tree            | ✓ |
| Random Forest            | ✓ |
| SVM                      | ✓ |
| XGBoost                  | ✓ |
| Cross Validation         | ✓ |
| Hyperparameter<br>Tuning | ✓ |
| Confusion Matrix         | ✓ |
| ROC-AUC                  | ✓ |
| Feature Importance       | ✓ |
| Explainable AI           | ✓ |
| Streamlit                | ✓ |
| End-to-end ML            | ✓ |

## Other excellent projects after this

Once they complete this one, I would give different groups projects such as:

1. **Employee Attrition Prediction** — classification
2. **Loan Approval Prediction** — classification
3. **House Price Prediction** — regression
4. **Customer Churn Prediction** — classification
5. **Used Car Price Prediction** — regression
6. **Credit Risk Prediction** — classification
7. **Medical Insurance Cost Prediction** — regression
8. **E-commerce Customer Purchase Prediction** — classification
9. **Retail Sales Forecasting** — regression/time series
10. **Fraud Detection** — imbalanced classification

**For teaching, I'd start with Student Placement Prediction, then give each student/group a different domain using the same ML pipeline.** This lets you teach one standardized project methodology while still giving them individual portfolio projects.

- 1) **Medical Insurance Cost Prediction** — regression
- 2) **E-commerce Customer Purchase Prediction** — classification
- 3) **Retail Sales Forecasting** — regression/time series